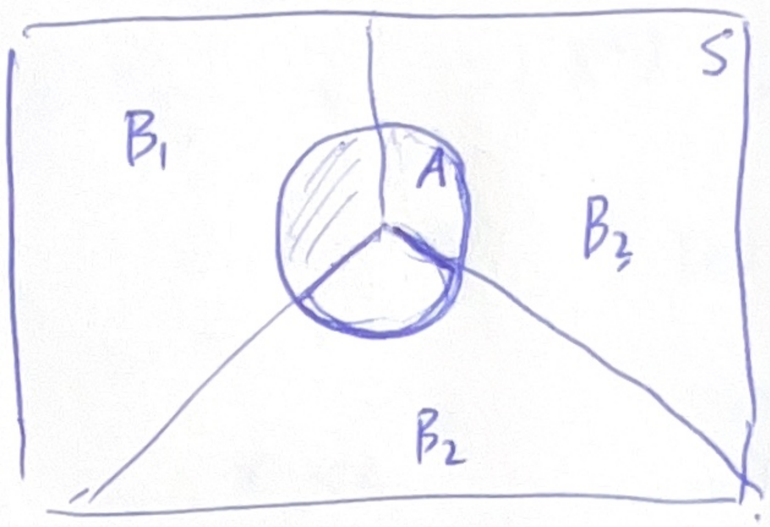
Review

Law of total probability

$P(A) = ?$



$$P(A) = P(A \cap B_1) + P(A \cap B_2) + P(A \cap B_3)$$

$$= P(B_1)P(A|B_1) + P(B_2)P(A|B_2) + P(B_3)P(A|B_3)$$

$$P(A \cap B) = P(A)P(B|A)$$
$$= P(B)P(A|B)$$

Baye's Rule

$P(A|B) = ?$ you know $P(B|A)$

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

$$= \frac{P(A \cap B)}{P(B)} = \frac{P(B|A)P(A)}{P(B)}$$

Complement Rule

$$P(A) = 1 - P(A')$$
$$P(A|B) = 1 - P(A'|B)$$

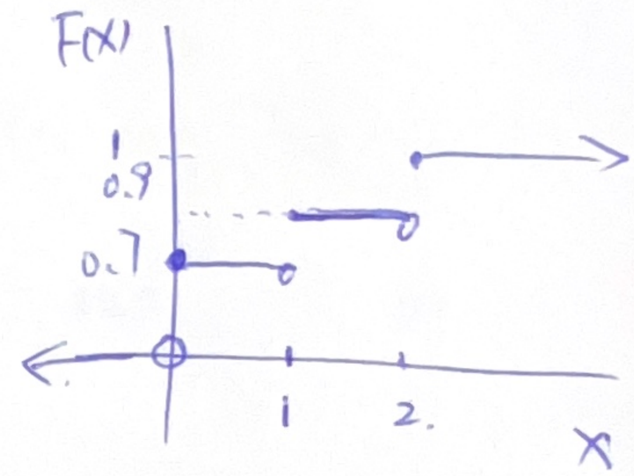
Theorem: Computational formula for variance.

The variance of a R.V can be found by

$$Var(X) = E(X^2) - [E(X)]^2 = E(X^2) - \mu^2.$$

Example:

$$F(x) = \begin{cases} 0 & x < 0 \\ 0.7 & 0 \le x < 1 \\ 0.9 & 1 \le x < 2 \\ 1 & 2 \le x \end{cases}$$



To find pmf.

$$f(x) = \begin{cases} 0.7 & x = 0. \\ 0.2 & x = 1 \\ 0.1 & x = 2. \end{cases}$$